



TomoMamba: a two-stage model with state-space cross-slice propagation for breast cancer diagnosis in digital breast tomosynthesis

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Abstract

Digital breast tomosynthesis (DBT) provides volumetric image stacks that improve lesion conspicuity compared with two-dimensional (2D) mammography, yet most automated methods process slices independently and discard the depth-wise continuity that underpins this advantage. We introduce TomoMamba, a two-stage deep learning framework for DBT screening and diagnosis. A dual-branch 2.5D network with multi-view fusion first triages each breast study as normal or suspicious, reaching a breast-level area under the receiver operating characteristic curve (AUC) of 0.97. Selective state-space (Mamba) models then propagate features along the depth axis at each spatial position through input-dependent state transitions; lesions are localized by an anchor-free CenterNet detector and classified from features cropped at the detected centers. On the public BCS-DBT dataset the cascade reaches a breast-level AUC of 0.93 (95% CI 0.85–0.99) on validation and 0.92 (95% CI 0.86–0.97) on the held-out test partition. On validation it loses two of 20 cancers at screening while forwarding 14% of breasts, so this cascade figure reflects the screening stage rather than benign-versus-cancer discrimination. In isolation, Stage 2 attains a patient-level AUC of 0.80 on validation and 0.62 on test, with lesion detection of 72.0% and 62.5% respectively; as the test classification interval spans chance, we present lesion localization as Stage 2's demonstrated contribution. The gated cross-slice residual is the only component whose removal produces a significant change (0.25 AUC, $p = 0.009$). To the best of the authors' knowledge, this is the first application of selective state-space models to inter-slice feature propagation in DBT.

Keywords Digital breast tomosynthesis · Breast cancer detection · State-space models · Anchor-free detection · Deep learning · Computer-aided diagnosis

1 Introduction

Breast cancer remains the most frequently diagnosed malignancy among women worldwide, with approximately 2.3 million new cases and 685,000 deaths reported annually [22]. Stage at detection is strongly associated with outcome: the five-year relative survival rate exceeds 99% for localized disease but declines sharply at advanced stages [20]. Digital breast tomosynthesis (DBT) has emerged as an important advance over conventional two-dimensional (2D) mammography by providing pseudo-three-dimensional (3D) volumetric imaging, which reduces tissue overlap and enhances lesion conspicuity, particularly in patients with dense breast tissue [5, 9].

Despite these clinical advantages, DBT introduces significant challenges for both radiologists and automated analysis systems. Each examination typically comprises dozens to hundreds of slices per view, increasing interpretation time and cognitive burden [3]. From a computational perspective, DBT represents a particularly demanding form of clinical imaging data: each study consists of multi-view, multi-slice volumes that must be analyzed jointly, which requires efficient volumetric feature extraction. Moreover, publicly available DBT datasets exhibit severe class imbalance, with normal cases outnumbering abnormal ones by a wide margin, which complicates model training [4]. Effective exploitation of cross-slice context is therefore essential for accurate lesion detection and characterization in this volumetric setting.

Recent studies have explored a variety of deep learning strategies for DBT analysis, including vision transformer-based slice-level classification with subsequent aggregation to volume-level predictions [12], self-supervised contrastive learning to mitigate class imbalance [7], and two-stage pipelines integrating classification and detection modules [1]. However, the majority of these approaches process slices independently, thereby overlooking the inter-slice continuity that underlies the diagnostic benefit of DBT. Methods that do explicitly model the depth dimension typically rely on full 3D convolutions, which are memory-intensive and difficult to optimize when annotated volumes number in the low hundreds. Beyond architectural constraints, a conceptual gap has also been highlighted: as noted by Zafari et al. [27], most breast imaging models perform a direct mapping from pixels to labels, bypassing the intermediate reasoning steps (detection, characterization, cross-view correlation, and evidence integration) that are fundamental to radiological decision making. This observation motivates stage-wise frameworks that more closely reflect clinical workflow.

State-space models (SSMs), particularly the Mamba architecture [10], have recently demonstrated strong capability in modeling long-range dependencies in sequential data with linear computational complexity. Unlike transformer-based approaches, whose self-attention scales quadratically with sequence length, Mamba enables efficient sequence modeling through selective, input-dependent state-space representations. In the context of DBT, this property is particularly advantageous because each spatial position across slices can be naturally represented as a one-dimensional sequence along the depth axis. While prior work has explored Mamba-based architectures for 2D mammography [2], their application to inter-slice feature propagation in 3D DBT remains largely unexplored. Newer variants such as Mamba-2 [6] have been proposed, but their efficiency gains are negligible for the short depth sequences used here (15 slices per volume), making the original Mamba architecture a more mature and stable choice in this setting.

In this work, we introduce **TomoMamba**, a two-stage deep learning framework for automated DBT screening and diagnosis. The main contributions of this study are:

1. **Clinical workflow-aligned two-stage design:** a screening-then-diagnosis pipeline in which the first stage efficiently classifies breast-level studies as normal or suspicious, and the second stage jointly discriminates benign from cancerous lesions while localizing them.
2. **Mamba-based cross-slice propagation for DBT:** to the best of the authors' knowledge, the first use of selective state-space models for inter-slice feature propagation in DBT. For each spatial position in the feature map, a 15-element depth sequence is processed by input-dependent state transitions that adapt to the content at each position, capturing volumetric context that slice-independent methods discard and that 3D convolutions obtain only at substantial memory cost.
3. **Region-of-interest (ROI) based classification from detected lesion centers:** rather than relying on global average pooling, which dilutes the classification signal across the full feature map, we crop 7×7 feature patches at CenterNet-detected lesion centers [28]. This provides the classifier with a direct gradient path to the relevant region, stabilizing joint detection and classification training on only 200 biopsied volumes, a regime where a pooling-only baseline oscillated between all-cancer and all-benign predictions without converging.
4. **Evaluation on all three official partitions with controlled baselines:** we evaluate the proposed framework on the public BCS-DBT dataset [4] with extensive ablation studies, controlled baseline comparisons under identical experimental conditions, and contextualization against published methods. Because BCS-DBT remains the only public DBT resource with lesion-level annotations, we position this work as a first controlled evaluation of state-space cross-slice propagation in DBT rather than as a clinically deployable system, and report performance on all three official partitions so that the limits of what 101 biopsied training patients can support are visible.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the proposed framework. Section 4 introduces the dataset and experimental setup. Section 5 presents results and discussion, and Section 6 concludes the paper.

2 Related work

Deep learning for DBT analysis has expanded rapidly since the release of the BCS-DBT dataset [4]. We organize prior work according to how each method handles the depth dimension of DBT volumes, since this is the axis along which our contribution lies.

Lesion detection The earliest public baseline for DBT lesion detection was introduced by Buda et al. [4], who combined a DenseNet backbone with YOLO-style detection heads operating on individual slices. Treating each slice independently, this approach achieved a per-breast sensitivity of 65% at two false positives on the official BCS-DBT test set. Building on this benchmark, the DBTex challenge [13] attracted eight competing teams and substantially advanced performance. The top-performing entry used an ensemble of 35

EfficientDet models trained on more than one million external mammography images and 400,000 DBT volumes, achieving a mean biopsied-lesion sensitivity of 95.7%. In a more constrained setting, Hossain et al. [11] reported 74.3% sensitivity at two false positives per image on the 985-patient DBTex Phase 1 subset through false-positive mining, but without explicit cross-slice modeling.

Slice-independent classification A substantial body of work formulates DBT classification by treating each slice or volume as an independent sample, without explicitly modeling inter-slice dependencies along the depth axis. Elshazli et al. [8] proposed a modified AlexNet architecture augmented with HSV color mapping for three-class slice-level classification, achieving 91.6% accuracy on a custom 80/20 split of BCS-DBT subsets of up to 600 patients. To address the pronounced class imbalance in DBT data, Du et al. [7] introduced a self-supervised contrastive pre-training framework (SIFT-DBT) that aggregates slice-level predictions into volume-level outputs, yielding an AUC of 0.9269 on a custom 7:1:2 split of the BCS-DBT training partition. Similarly, Yang et al. [24] combined shallow and deep feature representations with pseudo-color enhancement, reporting an image-level AUC of 0.822 on the official BCS-DBT split. Transformer-based approaches have also been explored; for instance, Kassis et al. [12] applied Swin Transformers to individual slices and incorporated post-processing that aggregates scores from neighboring slices, achieving a case-based AUC of 0.918 ± 0.014 on a proprietary 3,831-scan dataset from Soroka Medical Center. Although the post-processing in Yang et al. and Kassis et al. considers adjacent slices, the feature extraction itself remains strictly per-slice.

Depth-modeling approaches A smaller set of studies explicitly models inter-slice context within the learning architecture. Park et al. [18] proposed 3D-GMIC, a globally aware multiple-instance learning framework that processes entire DBT volumes while reducing GPU memory consumption by 78–90% compared with conventional 3D convolutional neural networks (CNNs). Trained on 85,526 NYU Langone patients and evaluated on BCS-DBT as an external test set, the method achieved an image-wise AUC of 0.848. Likewise, Lee et al. [14] introduced a TimeSformer-based architecture in which a CNN backbone extracts per-section features that are subsequently integrated through a transformer-based interaction module. Their model was trained, validated, and tested on 5,174, 1,000, and 655 four-view DBT studies, respectively, retrospectively collected from nine US institutions. It achieved a study-level AUC of 0.91 ± 0.03 , outperforming both 2D slice-wise baselines and full 3D convolutional models while requiring only 25% of the floating-point operations of the latter. Nonetheless, the self-attention mechanism in such models scales as $O(D^2)$ with the number of slices. A different strategy, that of Terrassin et al. [23], sidesteps explicit depth modeling by training a 2D CNN on full-field digital mammograms and applying it to DBT thick slices, achieving an AUC of 0.90 on the official BCS-DBT splits. Notably, none of these approaches jointly performs classification and lesion localization.

Two-stage classification and detection On 2D mammography, Soltani et al. [21] fine-tuned Mask R-CNN to segment breast lesions and then classified the segmented regions with pre-trained CNNs. The most closely related prior work is that of Alashban et al. [1], who propose a two-stage framework combining a modified VGG19 classifier for three-class categorization with a YOLOv5-CBAM detector for per-slice lesion localization, evaluated on

the official BCS-DBT partition. However, the two components operate independently: the classifier processes full images without leveraging localization cues from the detector, and neither module captures cross-slice dependencies. In contrast, our Stage 2 introduces three concrete differences. First, classification and detection are unified within a shared backbone combining ResNet-18, a feature pyramid network (FPN) and Mamba, so the classifier receives features that are already informed by the detection heatmap. Second, classification is performed on ROI-pooled features centered on detected lesion regions, providing a more focused and clinically relevant representation than full-image processing. Third, Mamba propagates features along the depth axis at each spatial position with $O(D)$ complexity, providing cross-slice context that is absent in slice-by-slice designs.

State-space models in medical imaging State-space models have recently emerged as computationally efficient alternatives to transformer-based architectures for sequence modeling. Unlike self-attention, their complexity scales linearly with sequence length, while input-dependent state transitions enable selective information retention and filtering at each step. The Mamba architecture [10] formalized this selective state-space framework and demonstrated strong performance on long-range sequential tasks.

In medical imaging, however, the adoption of SSM-based models has so far been largely limited to two-dimensional settings. Mammo-Mamba [2] introduced a hybrid SSM–transformer framework with a sequential mixture of experts for multi-view mammography classification, while U-Mamba [15] and ResMamba-ULite [16] extended SSMs to general medical image segmentation. More recently, Zafari et al. [25] proposed a hybrid CNN–VSSM architecture to model cross-view dependencies in multi-view mammography. A common characteristic across these approaches is that Mamba is applied over the spatial dimensions of 2D images. None of them models the depth dimension inherent in volumetric medical data. This gap is particularly important in DBT, where clinically relevant information is distributed across slices. In our approach, we explicitly model this dimension by treating per-slice features at each spatial location as a short depth-wise sequence. For each spatial position in the feature map, a 15-element sequence of slice-wise features is processed by Mamba from front to back through the breast volume, enabling efficient and structured cross-slice reasoning.

Sequence modeling in medical imaging without self-attention has followed two broad lines. The first uses fixed recurrence: convolutional LSTM and gated recurrent modules have been applied to slice stacks and cine sequences [17, 19]. The second is the structured state-space family described above, whose state update depends on the input. In conventional gated recurrence, the learned transition functions are shared across steps, whereas a selective state-space model makes its discretized state-space parameters input-dependent at each step. This distinction is the basis of our architectural claim. We adopt a bidirectional GRU as the closest transformer-free control and compare against it under a matched parameter budget and identical training conditions in Section 5.4, alongside a transformer variant [14] that isolates the cost of attention.

3 Materials and methods

The proposed TomoMamba framework consists of two sequential stages, as illustrated in Fig. 1. Stage 1 performs breast-level screening, classifying each study as normal or suspicious. Only studies classified as suspicious are passed to Stage 2, which performs joint benign-versus-cancer classification and lesion localization.

Stage 1: Screening with a 2.5D CNN and multi-view fusion Stage 1 performs breast-level binary classification (normal vs. suspicious) using a dual-branch 2.5D convolutional architecture with explicit multi-view fusion (Fig. 2).

Each branch processes one view type through a shared ResNet-18 backbone (pre-trained on ImageNet) while aggregating information across slices with a learnable attention mechanism. For a volume with D slices, the backbone extracts per-slice feature vectors $f_i \in \mathbb{R}^d$

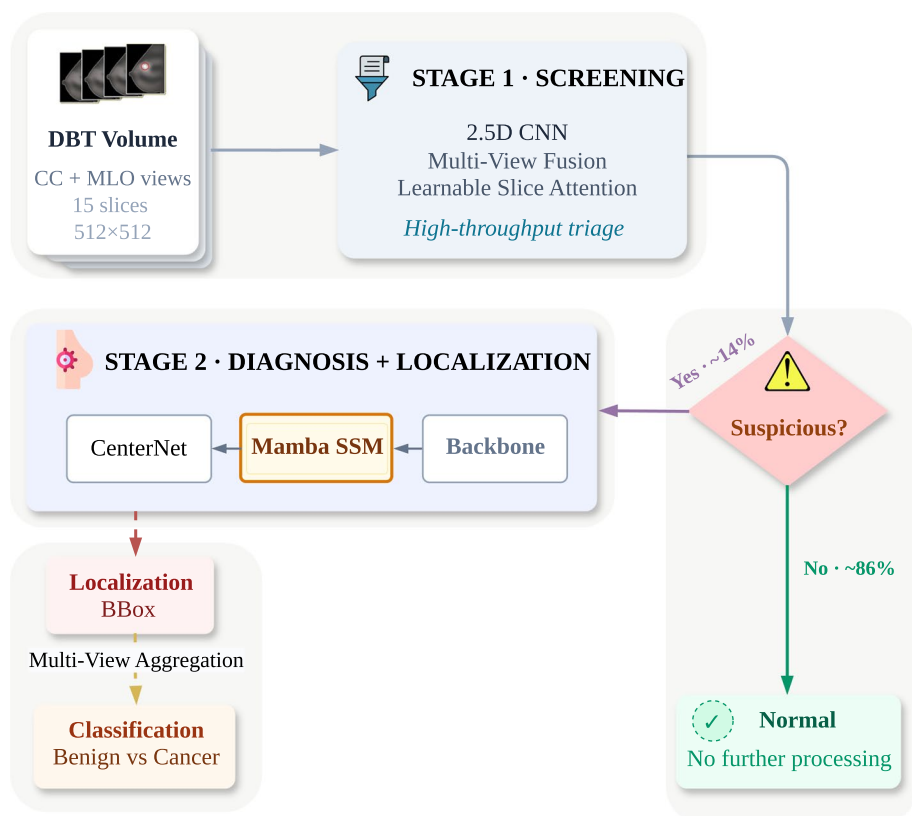


Fig. 1 Overview of the proposed TomoMamba framework. Stage 1 screens DBT studies as normal or suspicious using a 2.5D CNN with multi-view fusion. Suspicious cases are forwarded to Stage 2, which employs a Mamba-enhanced anchor-free detector for joint classification and lesion localization. The proportion shown in the schematic is illustrative only; the measured fraction of breasts forwarded from Stage 1 to Stage 2 at the operating threshold $\tau_{\text{scr}} = 0.139$ is 14% on validation (56 of 405) and 13% on test (98 of 738), as reported in Section 5.3 and Table 8

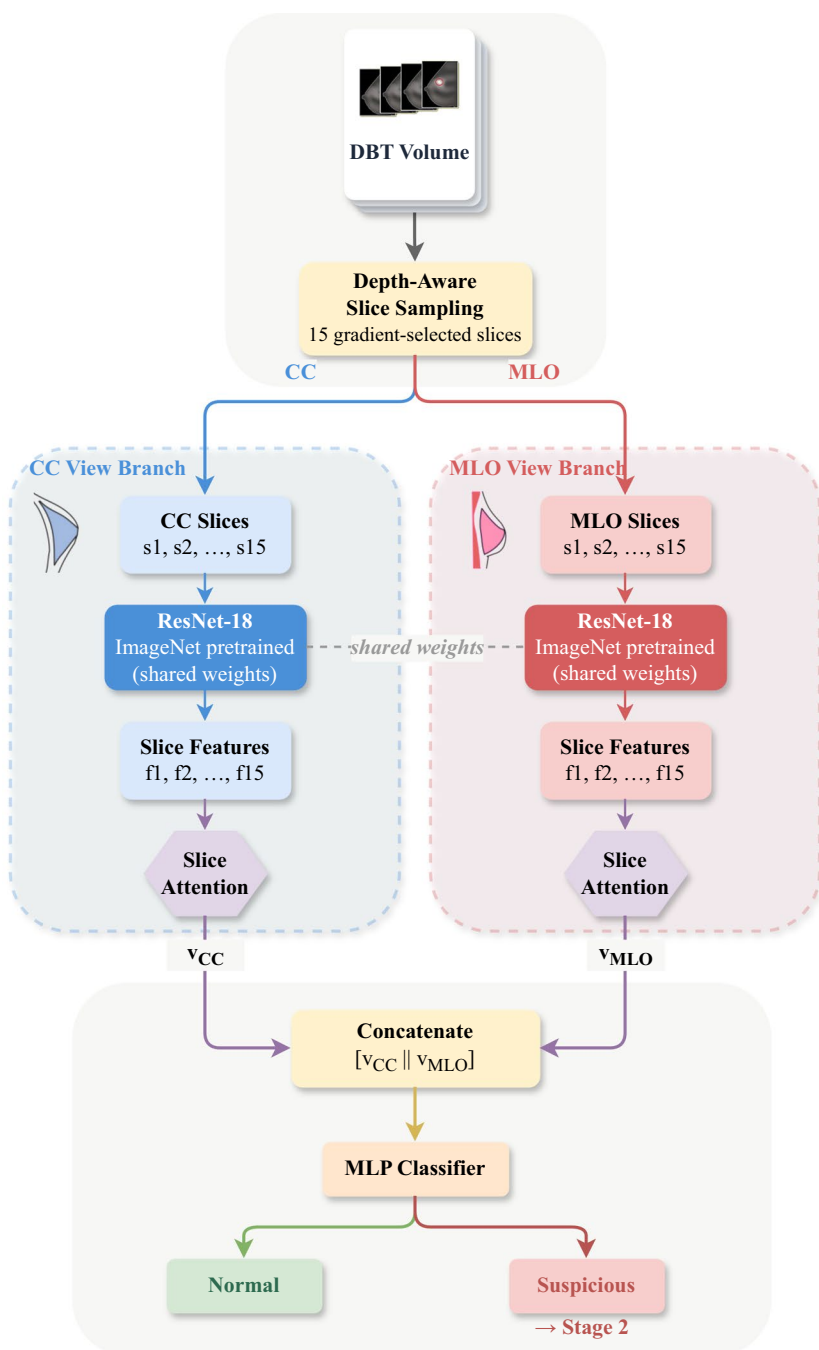


Fig. 2 Stage 1 architecture: A dual-branch 2.5D CNN processes CC and MLO views independently through shared convolutional layers with learnable slice attention. View-level features are fused and passed through a binary classifier

($i = 1, \dots, D$) via global average pooling of the final convolutional layer. The slice attention assigns importance weights:

$$\alpha_i = \frac{\exp(w^T f_i + b)}{\sum_{j=1}^D \exp(w^T f_j + b)}, \quad v = \sum_{i=1}^D \alpha_i f_i, \quad (1)$$

where $w \in \mathbb{R}^d$ and $b \in \mathbb{R}$ are learnable parameters, and $v \in \mathbb{R}^d$ is the view-level representation.

View-level representations from the CC and MLO branches are concatenated and classified via: $p(\text{suspicious} | x) = \sigma(\text{MLP}([v_{CC} || v_{MLO}]))$. Inverse-frequency class weighting ($\sim 7.78\times$ on the minority class) addresses the severe imbalance.

Stage 2: TomoMamba for joint classification and localization Suspicious studies are analyzed simultaneously for two tasks: benign-versus-cancer classification and lesion localization. The architecture consists of a ResNet-18 feature backbone with a lightweight FPN, Mamba-based cross-slice propagation modules, and CenterNet-style detection heads (Fig. 3).

For each slice $s \in \{1, \dots, D\}$, the backbone extracts multi-scale features from ResNet-18 layers 1–4, which are upsampled and concatenated via a lightweight FPN (FPN-Lite). A fusion convolution reduces the concatenated features to produce $F_s \in \mathbb{R}^{128 \times H' \times W'}$ (where $H' = W' = 96$ at stride 4 from a 384×384 input). Using a 2D backbone with shared weights across slices lets the model reuse pre-trained ImageNet features while keeping memory manageable.

Mamba cross-slice propagation The key novelty of Stage 2 is the application of Mamba SSMs [10] for cross-slice feature propagation. For each spatial position (h, w) in the feature map, we extract a depth sequence $z_{h,w} = [F_1[h, w], \dots, F_D[h, w]] \in \mathbb{R}^{D \times 128}$ and process it through the Mamba selective state-space model:

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \quad y_t = C_t h_t, \quad (2)$$

where $\bar{A}_t, \bar{B}_t$ are discretized, input-dependent matrices (Mamba's selective mechanism with $d_{\text{model}} = 128$, $d_{\text{state}} = 16$, $d_{\text{conv}} = 4$, expand factor = 2), C_t is the input-dependent output projection, $x_t = z_{h,w}[t]$ is the feature vector at depth step t , and h_t is the hidden state. This gives linear $O(D)$ complexity, against $O(D^2)$ for transformers.

The Mamba output is integrated through a gated residual connection:

$$g = \sigma(W_g[x || y]), \quad \tilde{F}_s[h, w] = F_s[h, w] + g \odot y_s, \quad (3)$$

where W_g is a learnable projection and σ is the sigmoid function, y_s denotes the Mamba output y_t of Eq. 2 at depth step $t = s$, and $\tilde{F}_s$ is the depth-enhanced feature map used by every downstream head. This gating mechanism allows the model to adaptively control how much cross-slice context to incorporate at each spatial position, preserving strong per-slice

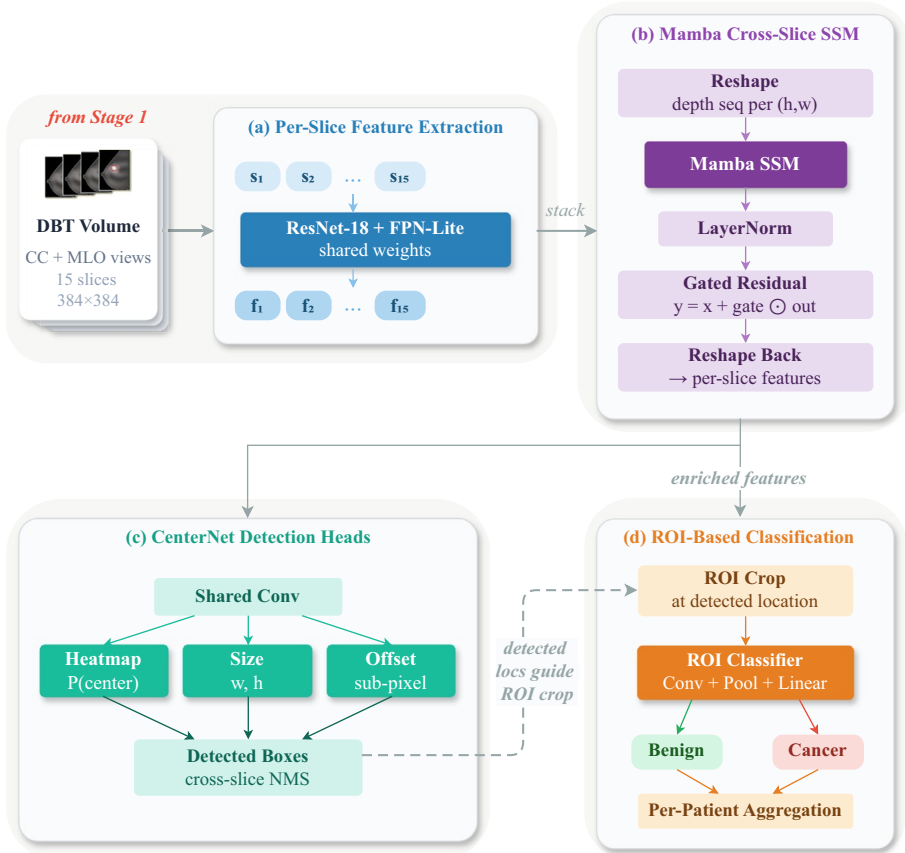


Fig. 3 Stage 2 TomoMamba architecture. ResNet-18 extracts per-slice features (F_s), Mamba processes depth sequences at each spatial position to produce enhanced features ($\tilde{F}_s$), and three CenterNet-style heads predict heatmaps, size maps, and offsets directly from $\tilde{F}_s$. ROI features are cropped from $\tilde{F}_s$ at the resulting lesion centers and passed to the classifier, so ROI pooling follows detection rather than preceding it

features while enriching them with depth information. A LayerNorm is applied before gating for training stability.

Figure 4 illustrates this mechanism. At position (h, w) , the feature map has 15 rows (slices) $\times$ 128 channels (panel a). A lesion activates strongly on 2–3 adjacent slices. Because $\tilde{B}_t$ is input-dependent, the contribution each slice makes to the hidden state varies with its content, so the resulting depth summary carries information about where signal was observed along the depth axis (panel b). The gated residual then makes per-channel keep/suppress decisions conditioned on this summary (panel c). Introspection across 72 validation volumes (13,167 lesion feature cells) shows that the gate is channel-selective rather than slice-selective. At lesion positions a mean of 40.0 of 128 channels fall outside the neutral band ($g > 0.7$ or $g < 0.3$), against 23.5 at background positions, a contrast of 1.70 $\times$ with Cohen's $d = 1.69$. The direction holds in 67 of 72 volumes individually. Table 1 reports the

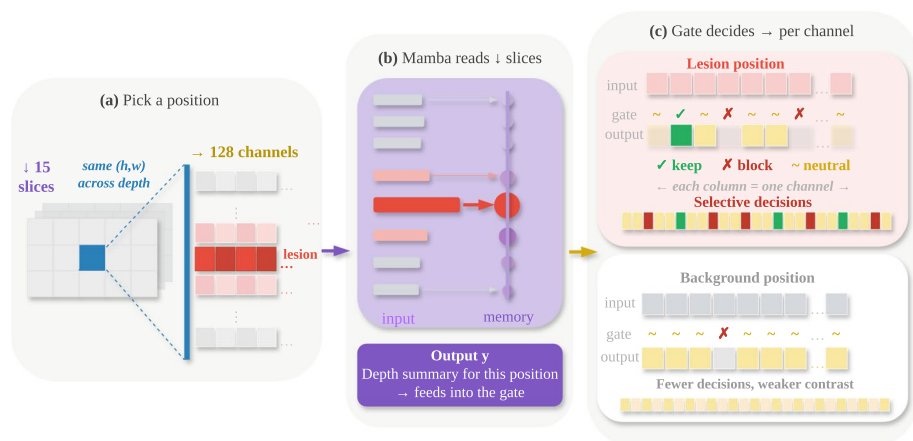


Fig. 4 Mamba cross-slice mechanism at a single spatial position, shown for a representative case. **(a)** A position is selected from the feature map; zooming in reveals 15 rows (slices) $\times$ 128 columns (channels), with the lesion slice showing strong activation. **(b)** Mamba reads the depth sequence slice by slice; because the state update is input-dependent, the amount each slice contributes to the state varies with its content (larger dot, larger contribution). **(c)** The gated residual uses Mamba's depth context to make per-channel keep/block decisions. At lesion positions roughly a third of channels fall outside the neutral band; at background positions fewer than a fifth do, so the gate acts at both kinds of position but discriminates more strongly at lesions. Gate statistics are quantified in Table 1 over 72 validation volumes and 13,167 lesion feature cells; lesion positions are feature cells inside a ground-truth box on the annotated slice, and background positions are cells at least four feature cells from every box, randomly subsampled to match the lesion count

full statistics together with a matched control in which Mamba is replaced by a bidirectional GRU.

CenterNet detection heads Three parallel heads operate on the Mamba-enhanced features $\tilde{F}_s$ through two shared convolutional layers ($128 \rightarrow 128 \rightarrow 64$ channels). The *heatmap head* predicts $\hat{Y} \in [0, 1]^{H' \times W'}$ with Gaussian ground-truth kernels centered at lesion positions, trained with modified focal loss ($\alpha = 2, \beta = 4$). A detection is scored against that slice under the depth-tolerance convention of [4]. The model receives only pixel data, with nothing in the input distinguishing the annotated slice from the other fourteen. The output bias is initialized to -2.19 (corresponding to a prior probability of ~ 0.1) to stabilize early training. The resulting heatmap highlights suspicious regions and is directly interpretable. The *size head* predicts normalized bounding-box width and height at each center via sigmoid activation, trained with L_1 loss computed only at ground-truth center locations. The *offset head* predicts sub-pixel center offsets ($\Delta x, \Delta y$) to compensate for the discretization error introduced by the stride-4 feature map, trained with L_1 loss at ground-truth center locations.

ROI-based classification Rather than global average pooling, which spreads the lesion signal over the entire feature map, classification operates on ROI-pooled features. Specifically, a 7×7 feature patch is bilinearly interpolated from $\tilde{F}_s$ at each detected lesion center with 30% context padding. The ROI classifier consists of two convolutional layers ($128 \rightarrow 256 \rightarrow 256$) with BatchNorm and Dropout2d (0.3), followed by adaptive average pooling and fully connected layers ($256 \rightarrow 64 \rightarrow 2$) with Dropout (0.7). A global average pooling fallback

Table 1 Gate activity at lesion positions compared with background positions, for TomoMamba and for a matched control in which Mamba is replaced by a bidirectional GRU. Statistics are computed over 72 validation volumes and 13,167 lesion feature cells. Lesion positions are feature cells inside a ground-truth box on the annotated slice; background positions are cells at least four feature cells from every box, randomly subsampled to match the lesion count. Ground-truth boxes are used, so these statistics do not depend on detector quality. Values are mean $\pm$ SD across cells; for the bidirectional GRU control only the lesion-to-background ratio and effect size are given

	Lesion	Background	Ratio (<i>d</i>)
Active channels (<i>g</i> > 0.7 or <i>g</i> < 0.3)	39.96 ± 10.01	4.48 ± 9.44	1.70× (1.69)
of which amplified (<i>g</i> > 0.7)	17.47 ± 5.64	0.05 ± 4.84	1.93× (1.60)
of which suppressed (<i>g</i> < 0.3)	22.49 ± 5.44	4.42 ± 5.27	1.56× (1.50)
Channel-wise SD	0.182 ± 0.002	0.143 ± 0.021	1.27× (1.83)
Gate value range (max – min)	0.799 ± 0.009	0.21 ± 0.056	1.11× (1.36)
Volumes with higher lesion selectivity	67 / 72		
<i>Bidirectional GRU control, same statistics</i>			
Active channels	—	—	1.68× (1.65)
Channel-wise SD	—	—	1.25× (1.65)
<i>Effect of removing the gated residual</i>			
With	PatAUC = 0.798		
Without	PatAUC = 0.548		
Difference	–0.250 (<i>p</i> = 0.009, DeLong)		

classifier is used when no detections are produced. Patient-level predictions are obtained by max-pooling cancer probabilities across all detected ROIs and views.

Multi-task loss The total loss is:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{hm}} \mathcal{L}_{\text{hm}} + \lambda_{\text{size}} \mathcal{L}_{\text{size}} + \lambda_{\text{off}} \mathcal{L}_{\text{off}} + \lambda_{\text{peak}} \mathcal{L}_{\text{peak}}, \quad (4)$$

where $\lambda_{\text{cls}} = 5.0$, $\lambda_{\text{hm}} = 1.0$, $\lambda_{\text{size}} = 0.3$, $\lambda_{\text{off}} = 1.0$, and $\lambda_{\text{peak}} = 0.3$. The classification loss is upweighted to ensure that the detection-dominant gradients (dense pixel-level supervision) do not suppress classification learning (single volume-level label). The peak loss $\mathcal{L}_{\text{peak}}$ encourages the heatmap to produce sharp activations at annotated lesion centers.

4 Dataset and experimental setup

We use the Breast Cancer Screening Digital Breast Tomosynthesis (BCS-DBT) dataset [4], the only publicly available DBT dataset with lesion-level bounding-box annotations [26], hosted on The Cancer Imaging Archive (TCIA). The dataset was collected at Duke University Hospital between August 2014 and January 2018 using a Hologic Selenia Dimensions system. No other public DBT dataset provides the combination of volumetric imaging data, biopsy-confirmed pathology labels, and spatial lesion annotations required for joint detection and classification research.

The dataset comprises 22,032 reconstructed DBT volumes from 5,610 screening studies of 5,060 patients (5,059 women; mean age 55 ± 11 years). Each study includes up to four anatomical views: left and right craniocaudal (LCC, RCC) and mediolateral oblique

(LMLO, RMLO). Studies are classified into four mutually exclusive categories based on clinical outcome: normal (91.4%), actionable (5.0%), biopsy-proven benign (2.0%), and biopsy-proven cancer (1.6%). The biopsy-proven categories were cross-matched with pathology reports from Duke's DEDUCE clinical data system. According to the dataset authors [4], two experienced radiologists (with 25 and 18 years of breast-imaging experience, respectively) annotated the biopsied cases. For each lesion, the annotator drew a rectangular bounding box on a slice showing the lesion and recorded the box coordinates, slice number, lesion class (benign or cancer), and architectural-distortion flag.

All volumes were acquired with an inter-slice spacing of 1 mm. Native in-plane dimensions range from approximately 1,800 to more than 3,000 pixels per axis depending on breast size, and the number of slices per volume varies with compressed breast thickness. This variability motivates our gradient-based slice selection (described below under preprocessing), which standardizes each volume to 15 representative slices.

The dataset provides official patient-level train, validation, and test splits, summarized in Table 2. We exclude the actionable category and retain only normal, biopsy-proven benign, and biopsy-proven cancer cases. The actionable label covers patients who were recalled but never biopsied, so no definitive diagnosis exists; excluding them follows established practice [1, 7].

The resulting class distribution is extreme. In the training set, normal cases outnumber cancer cases by more than 100:1. A single model trained on this distribution risks learning to predict "normal" for every input while still achieving more than 94% accuracy. This imbalance directly motivates our two-stage design: Stage 1 absorbs the screening burden by filtering out the large normal population at high sensitivity, allowing Stage 2 to operate on the smaller, more balanced suspicious subset where the benign-versus-cancer distinction is made.

Task formulation We reformulate the BCS-DBT categories into a clinically motivated two-stage pipeline:

- **Stage 1 (Screening):** binary classification of Normal versus Suspicious, where the Suspicious class combines Benign and Cancer cases (N_{train} : 1,803 Normal, 200 Suspicious volumes; N_{val} : 728 Normal, 75 Suspicious volumes). Normal volumes are subsampled from the full dataset. At training time, CC and MLO volumes belonging to the same breast are paired as a joint input. Normal volumes for Stage 1 training were subsampled at random from the full normal population without stratification, at approximately a 9:1 ratio to suspicious volumes, and the same subset was used for every experiment reported here. This ratio does not reflect the natural prevalence of the screening population, in which normals outnumber biopsied cases by more than 100:1. The reported filtering percentage should therefore be read as the behavior of the screening stage on

Table 2 BCS-DBT official dataset distribution. Numbers indicate patient counts

Split	Patients	Normal	Actionable	Benign	Cancer
Training	4,362	4,083 (93.6%)	178 (4.1%)	62 (1.4%)	39 (0.9%)
Validation	280	200 (71.4%)	40 (14.3%)	20 (7.1%)	20 (7.1%)
Test	418	298 (71.3%)	60 (14.4%)	30 (7.2%)	30 (7.2%)

a balanced training distribution rather than as a projected clinical workload. The end-to-end evaluation in Section 5.3 is conducted on the validation screening population of 405 breasts and repeated on the 738-breast test screening population. In each case every biopsied breast of the partition is included, together with the normal breasts of 150 of the 200 validation normal patients and of the full test normal population. Actionable cases, which were recalled but never biopsied, are excluded here because no pathology label exists; in clinical use they would be forwarded for diagnostic workup rather than assigned a benign-versus-cancer prediction.

- **Stage 2 (Diagnosis + Localization):** binary classification of Benign versus Cancer with simultaneous lesion localization, operating only on suspicious cases (N_{train} : 124 Benign, 76 Cancer volumes; N_{val} : 38 Benign, 37 Cancer volumes).

This decomposition mirrors clinical practice: Stage 1 performs high-throughput triage, maximizing sensitivity to avoid missed cancers, while Stage 2 performs detailed characterization on a smaller, more balanced subset. Stage 2 operates on a small training set of 200 volumes from 101 patients, which represents one of the fundamental challenges of this work.

Evaluation protocol For detection evaluation, we adopt the spatial matching criteria of [4] and [13]. A predicted bounding box is matched to a ground-truth box if two conditions are met: (i) the Euclidean distance between predicted and ground-truth box centers is less than $\max(\text{half-diagonal of GT box}, d_{\text{match}})$, where $d_{\text{match}} = 100$ pixels at the intermediate preprocessing resolution of 1024×1024 , corresponding to approximately 37.5 pixels at the model's operating resolution of 384×384 ; and (ii) the predicted center slice falls within $\pm 25\%$ of the total volume depth from the annotated center slice. Among candidates, the highest-scoring prediction is greedily assigned to each ground-truth box. We report detection rate (fraction of ground-truth lesions matched), $\text{Recall@IoU} \geq 0.25$ (R@0.25) computed over matched pairs, and mean IoU (mIoU) over matched pairs. Bounding boxes are expanded by a factor of 1.2 about their center during preprocessing so that the ROI classifier receives lesion margin in addition to lesion core; this expansion is applied identically to training and evaluation annotations, and all detection metrics are computed against the expanded boxes. Because the center-distance matching radius depends on the ground-truth box diagonal, the detection rate is accordingly more permissive than the same criterion applied to unexpanded annotations. Detection rate counts matched ground-truth lesions only, and Recall@IoU and mean IoU are computed over matched pairs, so unmatched predictions do not enter these three metrics; we report false positives per volume alongside them, and give the full free-response operating characteristic (FROC) analysis in Section 5.6. For classification, we report breast-level AUC for Stage 1 and patient-level AUC (PatAUC) for Stage 2, computed by max-pooling cancer probabilities across all detected ROIs and views belonging to each patient. In Stage 2, all but six patients across the three partitions have a single biopsied breast, so breast-level and patient-level metrics coincide except in those cases. Every test patient has exactly one, and no patient has biopsied breasts of differing classes. Table 3 gives the number of patients, breasts, volumes and lesion boxes at each stage and split. Patient-level sensitivity, specificity, and Youden's J index ($J = \text{Sens} + \text{Spec} - 1$) are reported at the standard classification threshold of $\tau_{\text{cls}} = 0.50$ on the max-pooled cancer probability.

Table 3 Evaluation units at each stage and split. Breasts are counted once per study, so the breast column carries study multiplicity and no separate study column is needed; the 405 validation and 738 test Stage 1 entries are 341 and 660 distinct breasts. Each volume is one view of one breast, so the volume column is also the view count. Stage 1 normal training volumes were subsampled, giving 1.4 volumes per breast entry, whereas the validation and test rows are complete two-view breasts. Stage 2 uses biopsied cases only, and six patients (five training, one validation) have two biopsied breasts of the same class, which is where patient-level and breast-level metrics differ

	Split	Patients	Breasts	Volumes	Lesion boxes
Stage 1	Train	—	—	2,003	—
	Validation	—	405	803	—
	Test	—	738	1,477	—
Stage 2	Train	101	106	200	224
	Validation	40	41	75	75
	Test	60	60	121	136

Data preprocessing Raw DBT volumes require careful preprocessing. Our pipeline follows the official BCS-DBT functions [4] with the following steps: (i) DICOM decompression using the pylibjpeg handler; (ii) window/level adjustment using DICOM header parameters to normalize pixel intensities; (iii) laterality correction by comparing edge intensity sums; (iv) gradient-based slice selection to retain 15 representative slices per volume, where each slice is scored by a weighted combination of Sobel edge magnitude and Laplacian variance ($0.6 \cdot \text{grad} + 0.4 \cdot \text{lap}/1000$), prioritizing slices with strong tissue boundaries and diagnostic content; for biopsied cases (Stage 2), slices carrying a radiologist bounding box are retained before the remaining slots are filled by gradient rank. This rule is applied identically at training, validation and test time. The retained slice is not additional information: it is present in the acquired volume and would be processed in full were it not for the memory constraint that forces downsampling from 50–100 slices to 15. It is retained because each lesion was annotated on exactly one slice, so a selection procedure that discards it leaves no ground truth against which a detection could be matched. The model receives only pixel data, with nothing in the input distinguishing the retained slice from the other fourteen. Selected slices are then sorted by original depth index, preserving anatomical order. Section 5.8 reports the empirical consequences of this choice, including results under a fully annotation-independent selection procedure; (v) percentile-based intensity normalization (1st–99th percentile clipping and rescaling to $[0, 1]$); and (vi) spatial resizing to 512×512 for Stage 1 and 384×384 for Stage 2. All preprocessed volumes are saved as NumPy arrays for efficient loading.

Implementation details Stage 1 uses the Adam optimizer with learning rate 10^{-4} , batch size 4, and cosine annealing over 50 epochs. Class weights are set to $[0.564, 4.390]$. Early stopping (patience 15) monitors validation AUC. The screening threshold $\tau_{\text{scr}} = 0.139$ was selected on a validation sweep targeting approximately 90% sensitivity; over the full 405-breast index it yields 0.878. Stage 2 uses AdamW with learning rate 3×10^{-4} , weight decay 5×10^{-3} , and batch size 4. ReduceLROnPlateau (patience 10, factor 0.5) monitors PatAUC. Mixed-precision training (FP16) is used throughout. Data augmentation includes horizontal flip, rotation ($\pm 15^\circ$), brightness/contrast jitter, Gaussian noise, and copy-paste lesion augmentation (whereby annotated lesion patches are composited onto other training volumes to synthetically increase lesion diversity). Concretely, each annotated lesion is extracted with a surrounding margin to form a crop pool; during training a crop is drawn

with probability 0.5 and alpha-blended (weight 0.7) onto a randomly chosen slice of a different volume that carries no annotation on that slice, and a matching bounding box is added for the pasted lesion. Like every augmentation listed here, copy-paste is applied only to the training partition; validation and test volumes are never augmented, so no synthesized lesion can enter an evaluation. Training runs for up to 200 epochs with early stopping on a composite PatAUC-plus-detection-recall score. The detection threshold $\tau_{\text{det}} = 0.16$ is selected via a validation sweep; the classification threshold $\tau_{\text{cls}} = 0.50$ operates on the patient-level max-pooled cancer probability. All experiments were conducted on a single NVIDIA RTX 4090 Laptop GPU (16 GB) with PyTorch 2.8 and CUDA 12.8.

5 Results and discussion

5.1 Main results

Table 4 summarizes the performance of both stages on the BCS-DBT validation set. Stage 1 achieves high screening performance (AUC = 0.9696), identifying 36 of 41 suspicious breasts (87.8%) while correctly filtering 94.5% of normal breasts. Stage 2 correctly classifies 29 of 40 patients (15 true positives, 14 true negatives, 6 false positives, 5 false negatives) and localizes 54 of 75 ground-truth lesion boxes (72.0% detection rate).

Figure 5 shows detection and heatmap outputs for five cancer and five benign validation cases. Predicted boxes align with the radiologist annotations across a range of lesion sizes and tissue densities, and heatmap peaks concentrate on lesion centers, not on surrounding tissue. Benign and malignant findings produce spatially similar activations, which is consistent with the classification signal coming from the ROI-pooled features at the detected center and not from heatmap intensity. The corresponding ROC curves for both stages are shown in Fig. 6. The gap between detection rate (72.0%) and R@0.25 (60.0%) reflects CenterNet's stride-4 resolution: the model finds the right location but produces approximate box dimensions. This is acceptable because the ROI classifier crops features at the detected center, not at the box boundaries, so center accuracy matters more than box precision. Under the stricter Recall@IoU ≥ 0.50 criterion the figure is 0.347, with a mean IoU of 0.475 over matched pairs, which is consistent with a detector that localizes lesion centers accurately while sizing boxes only approximately. The effect of detection-threshold selection on classification quality is examined in Section 5.5. The two-stage split lets Stage 1 optimize for sensitivity independently, while Stage 2 operates on a roughly 10 $\times$ smaller, more balanced subset with interpretable heatmap output.

Table 4 Main results on the BCS-DBT validation set. Stage 1 operates at screening threshold $\tau_{\text{scr}} = 0.139$; Stage 2 at detection threshold $\tau_{\text{det}} = 0.16$

	Task	AUC	Acc	Sens	Spec	R@0.25	mIoU	Det. Rate
Stage 1	Normal vs. Susp.	0.9696	0.938	0.878	0.945	—	—	—
Stage 2	Benign vs. Cancer	0.798	0.725	0.75	0.70	0.60	0.48	54/75 (72.0%)

Stage 1 AUC is breast-level; Stage 2 AUC is patient-level, obtained by max-pooling the cancer probability over a patient's views. The two levels coincide for Stage 2 wherever a patient has a single biopsied breast, which is every test patient, all but one validation patient and all but five training patients (Table 3); the Stage 2 figures may therefore be read at either level, and are labeled patient-level throughout for consistency

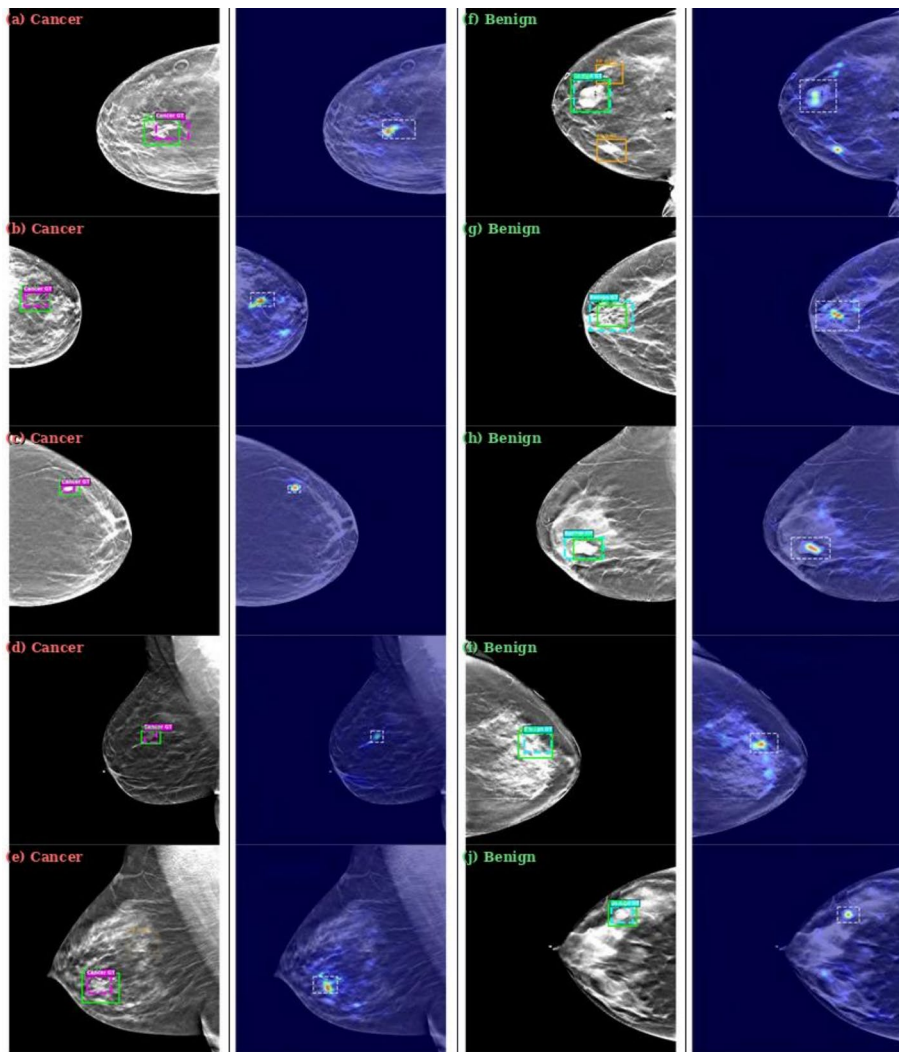


Fig. 5 Qualitative detection and heatmap results on the BCS-DBT validation set. Each case shows the GT+prediction overlay (left) and the corresponding CenterNet heatmap activation (right). Ground-truth boxes are dashed (magenta for cancer, cyan for benign); matched detections are solid green. (a)–(e) Cancer cases correctly detected and classified across different views (CC, MLO) and lesion sizes, with focused heatmap peaks at lesion centers. (f)–(j) Benign cases correctly localized and classified as non-malignant, demonstrating the model’s ability to distinguish benign from malignant findings

5.2 Generalization and stability

The results in Table 4 are computed on the validation partition, which was also used to select the model checkpoint and the operating thresholds τ_{scr} and τ_{det} ; τ_{cls} is fixed at the conventional value of 0.50 and was not tuned. To establish how much of that performance reflects discrimination and not selection, we report four further analyses: performance on

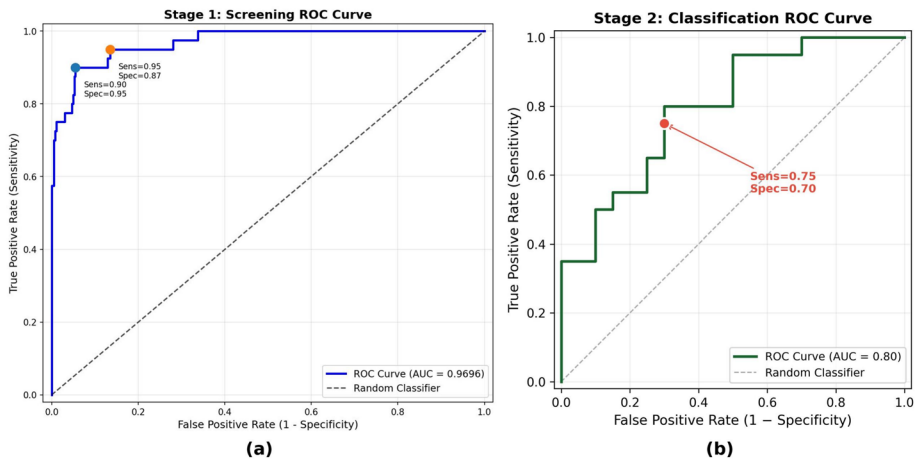


Fig. 6 ROC curves for (a) Stage 1 screening (Normal vs. Suspicious, AUC = 0.9696) and (b) Stage 2 classification (Benign vs. Cancer, PatAUC = 0.80). The two marked points on (a) are the curve positions nearest 90% and 95% sensitivity; the operating threshold $\tau_{scr} = 0.139$ reported in Table 4 lies between them, at sensitivity 0.878 and specificity 0.945. The marked point on (b) is the operating threshold $\tau_{cls} = 0.50$

Table 5 Stage 2 performance across the three official BCS-DBT partitions. Published checkpoint, $\tau_{det} = 0.16$, $\tau_{cls} = 0.50$. Intervals on AUC are bootstrap percentile at 95%; intervals on sensitivity, specificity and detection rate are Wilson score intervals at 95%. Detection rates are 204/224, 54/75 and 85/136 matched lesions respectively; the detection interval treats lesions as independent and so does not account for within-patient clustering

Partition	Patients	PatAUC (95% CI)	Sens	Spec	Det. Rate
Train	101	0.7713 [0.671, 0.864]	0.846 [0.703, 0.928]	0.597 [0.473, 0.710]	0.911 [0.866, 0.941]
Validation	40	0.7975 [0.645, 0.920]	0.750 [0.531, 0.888]	0.700 [0.481, 0.855]	0.720 [0.610, 0.809]
Test	60	0.6178 [0.469, 0.760]	0.700 [0.521, 0.833]	0.367 [0.219, 0.545]	0.625 [0.541, 0.702]

the held-out test partition, performance on the training partition, variability across random initializations, and grouped patient-level cross-validation.

The figures in this subsection, and in Tables 4, 5 and 9, use the annotation-guided slice selection described in Section 4. Because radiologist annotations are not available at inference, the annotation-independent evaluation in Section 5.8 is the deployment-relevant one; patient-level classification is essentially unchanged under it, and we regard those results as the primary indication of how the model would behave in practice.

Performance across partitions Table 5 reports Stage 2 on all three official BCS-DBT partitions using the published checkpoint at $\tau_{det} = 0.16$. Patient-level AUC on the untouched test partition is 0.6178 (95% CI [0.469, 0.760]), against 0.7975 on validation. Detection is more stable, falling from 72.0% to 62.5%. The classification drop is large, from 0.80 to 0.62, and we read it conservatively: on data not used for any selection the benign-versus-cancer signal is weak, its confidence interval includes chance, and we therefore make no diagnostic or

clinical claim for Stage 2 classification in isolation. The stronger and more stable quantity is lesion localization, which we present as Stage 2's demonstrated contribution.

Training-partition AUC (0.7713) is *lower* than validation AUC (0.7975): the model does not fit its training patients better than unseen ones, which rules out memorization of the training set as the source of the validation-to-test difference. This does not by itself rule out optimism from selecting on the validation partition, which we quantify separately below. Detection rate on the training partition is 91.1%, as expected for lesions seen during training; the relevant observation is the absence of any corresponding gap in classification.

Variability across random initializations We retrained Stage 2 five times on the fixed published split, varying only the random seed. The four additional runs used an epoch cap of 25 with early-stopping patience 8, against 200 and 20 for the published run. Every seed reached its best epoch between 6 and 18, so the shorter cap cannot have truncated any of them. Mean patient-level AUC is 0.6897 ± 0.0627 and mean detection rate is 0.6480 ± 0.1537 (Table 6).

The published checkpoint corresponds to the highest of the five seeds. A paired bootstrap comparison of each remaining seed against it gives p between 0.09 and 0.30, so no seed is statistically distinguishable from any other at this sample size.

Discrimination is considerably more stable across initializations than the operating point derived from it. Sensitivity and specificity at the fixed threshold $\tau_{\text{cls}} = 0.50$ vary from 0.750 to 1.000 and from 0.000 to 0.700 respectively across the same five runs, which reflects run-to-run differences in probability calibration rather than in ranking.

The five-seed mean of 0.6897 predicts the test-partition result (0.6178) much better than the published peak of 0.7975 does. Taken together with the absence of a train-validation gap, this points to selection on the validation partition, rather than memorization of the training data, as the principal source of the validation-to-test difference. We note that the difference between the two partitions is not itself statistically significant at these sample sizes.

Cross-validation As a further check that discrimination is not specific to the official split, we performed three-fold grouped patient-level cross-validation over the pooled training and validation partitions (275 volumes, 141 patients, StratifiedGroupKFold). Each fold was evaluated at the same $\tau_{\text{det}} = 0.16$ used elsewhere in the paper. Patient-level AUC across folds is 0.7145 ± 0.0873 . We report AUC because it is threshold-independent. The fixed $\tau_{\text{cls}} = 0.50$ is not calibrated per fold, so fold-to-fold variability at that threshold is a matter of calibration, not of discrimination, a point we return to in Section 5.11.

Summary Discrimination on the untouched test partition is weaker than on validation, and its confidence interval includes 0.50, so we do not claim that benign-versus-cancer discrimination is resolved at this sample size. Detection and localization, by contrast, transfer with only modest degradation across all three partitions. We present Stage 2's contribution

Table 6 Stage 2 variability across five random initializations on the fixed published split, evaluated on the validation partition at $\tau_{\text{det}} = 0.16$

Metric	Mean $\pm$ SD	Range
Patient-level AUC	0.6897 ± 0.0627	0.6475–0.7975
Detection rate	0.6480 ± 0.1537	0.400–0.773

primarily as lesion localization within a screening cascade, and treat the patient-level classification figure as an upper estimate obtained on the partition used for model selection.

5.3 End-to-end cascade evaluation

Stages 1 and 2 are evaluated independently above, which does not establish the behavior of the deployed system: cases missed by Stage 1 never reach Stage 2. The cascade is therefore evaluated over the complete validation screening population of 405 breasts (20 cancer, 21 biopsy-proven benign, 364 normal), so that Stage 2 receives only breasts forwarded by Stage 1. Its normal component derives from the 150 of the 200 validation normal patients who were preprocessed for Stage 1, since the cascade reuses that index. Breast entries are counted per study, so these 150 patients contribute 364 breast-studies (see Table 3). The test evaluation reported below uses that partition's *full* normal population of 298 patients and is therefore the complete-screening-population result; the validation figures are given alongside it for comparability with Table 4.

At the pre-selected screening threshold $\tau_{\text{scr}} = 0.139$, Stage 1 forwards 56 of 405 breasts, a workload reduction of 86.2%, and two of 20 cancer breasts (10.0%) and three of 21 benign breasts (14.3%) are lost at this step. Clinically this is the cascade's most consequential failure mode: roughly one cancer in ten is not forwarded to Stage 2 and would be missed if the cascade were used as a standalone filter. An operating point this aggressive is therefore appropriate only as a reader-support tool used alongside, not in place of, radiologist interpretation. The cancer-retention column of Table 7 makes the trade-off explicit: a more conservative, higher-retention threshold reduces the number of missed cancers only by forwarding more breasts, and we report the full curve so that this cost is visible rather than hidden behind a single operating point. Chaining the two stages gives a breast-level AUC of 0.9347 (95% CI [0.852, 0.993]) for cancer versus all other categories. At $\tau_{\text{cls}} = 0.50$ the sensitivity is 0.700 and the specificity 0.984, corresponding to 14 true positives, six false negatives, six false positives and 379 true negatives. A soft cascade that multiplies the probabilities of the two stages rather than thresholding Stage 1 gives 0.9206. Stage 1 forwards whole breasts and does not alter Stage 2's detection behavior, so lesion detection on forwarded breasts is 49 of the 67 reachable boxes (73.1%), marginally above the 72.0% obtained over all 75 boxes without screening. The end-to-end lesion sensitivity is lower than either figure, because lesions in breasts that are not forwarded can never be matched. Eight of the 75 annotated validation lesions lie in the eight biopsied volumes that are filtered at screening, and those volumes belong to the five breasts above; the end-to-end lesion sensitivity is therefore 49 of 75, or 65.3%. We report both, since the conditional figure characterizes the detector and the unconditional figure characterizes the deployed system.

Table 7 reports the screening-threshold trade-off. Cancer retention is flat at two lost cancers for $\tau_{\text{scr}} \leq 0.139$ and degrades above it, with five cancers lost by $\tau_{\text{scr}} = 0.30$. End-to-end AUC peaks at the pre-selected value.

Table 7 Screening-threshold trade-off for the complete cascade on the validation screening population (405 breasts). Lower thresholds retain more cancers but forward more breasts to Stage 2

τ_{scr}	Breasts forwarded	Cancers lost	End-to-end AUC
0.05	80	2	0.9239
0.10	63	2	0.9300
0.139	56	2	0.9347
0.20	50	4	0.8818
0.30	42	5	0.8599

The validation figures above use the partition on which τ_{scr} and τ_{det} were selected, and therefore carry the same optimism as the Stage 2 figures in Table 4; the test figures below do not.

We repeated the same evaluation on the test partition (738 breasts: 30 cancer, 30 biopsy-proven benign, 678 normal). Table 8 compares the two partitions. The same pattern holds for lesion detection. Eighteen of the 136 annotated test lesions lie in the 14 biopsied volumes filtered at screening, so the detector matches 80 of the 118 reachable boxes (67.8%); the end-to-end lesion sensitivity is 80 of 136, or 58.8%. The screening stage and the cascade as a whole transfer with little degradation, in contrast to Stage 2 evaluated in isolation, which falls from 0.7975 to 0.6178 (Table 5). We attribute this difference to composition, not to any discrepancy between the two evaluations. Of the 738 test breasts, 678 are normal, so the cascade AUC is dominated by the normal-versus-suspicious separation that Stage 1 performs well, and the high specificity follows from the same imbalance. Stage 2 alone, by contrast, is evaluated only on biopsied cases, where the benign-versus-cancer distinction constitutes the entire task. The high cascade AUC (0.92 on test) therefore reflects the normal-versus-suspicious separation performed by Stage 1 and must not be read as evidence that the Mamba module discriminates benign from cancer strongly. That far harder distinction is the Stage 2 figure of 0.62 on test, not the cascade figure, and the two should not be conflated. The soft cascade attains 0.9380 on the test partition, slightly above the hard cascade and reversing the ordering observed on validation. The confidence intervals overlap substantially, and we do not claim a preference between the two.

5.4 Controlled baseline comparison

Direct comparison with published methods is confounded by differences in task definitions, data splits, subset sizes, and external data usage (see Section 5.10). To isolate the effect of each proposed component, we therefore train and evaluate four baseline architectures under identical conditions: the same data split, preprocessing pipeline, augmentation strategy, optimizer settings, and evaluation protocol (Table 9). The only variable is the model architecture.

The ResNet-18 classifier (PatAUC = 0.675) never converged to a stable operating point, oscillating between all-cancer and all-benign predictions across epochs. Adding CenterNet

Table 8 End-to-end cascade on both partitions at $\tau_{\text{scr}} = 0.139$. Stage 2 evaluated in isolation is shown for comparison; it operates only on biopsied cases, whereas the cascade operates on the full screening population. Stage 1 figures here are recomputed by the end-to-end evaluation script over the full breast index; on validation it reproduces the specificity of Table 4 exactly and its AUC to within 0.001, and reproduces its sensitivity exactly

	Validation	Test
Breasts (cancer / benign / normal)	405 (20 / 21 / 364)	738 (30 / 30 / 678)
Stage 1 AUC (breast-level)	0.9705	0.9696
sensitivity / specificity	0.878 / 0.945	0.883 / 0.934
Cancers lost at screening	2 / 20 (10.0%)	3 / 30 (10.0%)
Benign lost at screening	3 / 21 (14.3%)	4 / 30 (13.3%)
Breasts forwarded	56 (86.2% reduction)	98 (86.7% reduction)
End-to-end AUC	0.9347 [0.852, 0.993]	0.9210 [0.859, 0.974]
sensitivity / specificity	0.700 / 0.984	0.667 / 0.965
end-to-end lesion sensitivity	49/75 (65.3%)	80/136 (58.8%)
Stage 2 alone (biopsied only)	0.7975	0.6178

detection heads immediately stabilizes training and raises PatAUC to 0.748, which is consistent with the importance of spatial grounding for classification at this data scale.

On the validation partition the cross-slice architectures are not separable by patient-level AUC. The bidirectional GRU reaches 0.808, TomoMamba 0.798 and the transformer variant 0.660. Paired DeLong tests give $p = 0.90$ and $p = 0.16$ respectively against TomoMamba, so we make no claim of superior discrimination on this partition. Two other differences are informative. The first is localization. A detection rate is meaningful only alongside the false-positive rate at which it is measured, so each baseline is compared against TomoMamba's own free-response curve at that baseline's operating point (Section 5.6). On the validation partition TomoMamba leads all three. It reaches 72.5% against the transformer variant's 65.3% at 1.57 false positives per volume, 58.9% against the bidirectional GRU's 54.7% at 0.84 (a margin of four points), and 65.9% against the slice-independent baseline's 64.0% at 1.01 (roughly one lesion). The picture on the test partition is mixed. TomoMamba leads the slice-independent baseline (67.6% against 61.8% at 1.73) and the transformer variant (68.1% against 64.0% at 1.81), but not the bidirectional GRU, which reaches 57.4% at 0.82 false positives per volume where TomoMamba reaches 55.6%. We restrict the localization claim to the validation partition. The second is transfer. On the untouched test partition TomoMamba attains the highest patient-level AUC of the configurations evaluated (0.618) and the lowest Brier score (0.242). The bidirectional GRU, the slice-independent baseline and the ResNet-18 classifier reach 0.493, 0.476 and 0.461 respectively, and the transformer variant reaches 0.551. None of these pairwise AUC differences reaches significance at 60

Table 9 Controlled baseline comparison for Stage 2 at $\tau_{\text{det}} = 0.16$ on both the validation and the held-out test partition. All models were trained on identical data, preprocessing and hyperparameters, each with independent early stopping; only the cross-slice module differs. Indented rows add the named module to the row above; all use the same ResNet-18 backbone. Confidence intervals are bootstrap percentile at 95%; p values are paired DeLong tests of the AUC difference against TomoMamba. J_{max} is the maximum Youden's J on each model's own ROC curve, with the attaining threshold in parentheses. Det. Rate counts matched ground-truth lesions; FP/vol counts unmatched detections per volume

Model	PatAUC (95% CI)	p	$J_{\text{max}} (\tau^*)$	Det. Rate	FP/vol	Brier
<i>Validation partition (40 patients, 75 lesion boxes)</i>						
ResNet-18 classifier ^a	0.675 [0.497, 0.835]	0.233	0.350 (0.510)	—	—	0.250
+ CenterNet ^b	0.748 [0.578, 0.897]	0.518	0.450 (0.624)	48/75 (64.0%)	1.01	0.222
+ BiGRU ^c	0.808 [0.651, 0.934]	0.898	0.550 (0.562)	41/75 (54.7%)	0.84	0.220
+ Transformer ^d	0.660 [0.481, 0.830]	0.156	0.400 (0.556)	49/75 (65.3%)	1.57	0.245
TomoMamba (Ours)	0.798 [0.645, 0.920]	—	0.500 (0.498)	54/75 (72.0%)	1.47	0.230
<i>Test partition (60 patients, 136 lesion boxes)</i>						
ResNet-18 classifier	0.461 [0.311, 0.612]	0.104	0.100 (0.512)	—	—	0.250
+ CenterNet	0.476 [0.326, 0.628]	0.119	0.133 (0.596)	84/136 (61.8%)	1.73	0.273
+ BiGRU	0.493 [0.343, 0.645]	0.147	0.133 (0.690)	78/136 (57.4%)	0.82	0.269
+ Transformer	0.551 [0.402, 0.695]	0.417	0.200 (0.625)	87/136 (64.0%)	1.81	0.253
TomoMamba (Ours)	0.618 [0.469, 0.760]	—	0.267 (0.557)	85/136 (62.5%)	1.34	0.242

^a Global average pooling on ResNet-18 features, followed by a linear classifier. No detection heads. Training was unstable, oscillating between predicting all-cancer and all-benign

^b CenterNet detection and classification, each slice processed independently (no cross-slice context)

^c Bidirectional GRU replaces Mamba for cross-slice propagation; architecture otherwise identical

^d A single-layer transformer encoder with a learned positional embedding replaces Mamba; architecture otherwise identical. The positional embedding is included because self-attention is permutation-invariant and would otherwise receive no depth ordering at all

patients (p between 0.10 and 0.42), so the ordering is descriptive. The Brier scores admit a fixed reference point: on a partition with equal numbers of benign and cancer patients, predicting 0.5 for every patient scores 0.25. TomoMamba is the only configuration below that value; the bidirectional GRU (0.269) and the slice-independent baseline (0.273) are above it, so their probability estimates on unseen data are less informative than a constant prediction. On the validation partition the ordering differs.

Youden's J at the fixed threshold $\tau_{\text{cls}} = 0.50$ is not used to support architectural claims here, because at a fixed threshold it conflates discrimination with calibration. The bidirectional GRU illustrates the point. Its J at $\tau_{\text{cls}} = 0.50$ is 0.05, but at its own optimal threshold it reaches 0.550, the highest J_{max} of any configuration on validation. Its low fixed-threshold figure is a calibration artifact, not evidence that it cannot discriminate. On the test partition the ordering reverses, with TomoMamba attaining the highest J_{max} at 0.267 against 0.133. No J comparison between any two of these architectures reaches significance on either partition; with only 40 and 60 patients, respectively, the bootstrap intervals on J span close to the full range of the statistic.

The localization differences on the validation partition follow from how each model uses depth. TomoMamba and the bidirectional GRU carry the same gated residual of (3), so the gate itself is not what distinguishes them; what differs is the depth summary the gate receives. Mamba's state transitions are input-dependent, allowing the depth summary to adapt to the content observed at each spatial position, whereas the BiGRU uses fixed learned recurrent transitions across inputs. The gated residual then translates that summary into per-channel feature decisions, as quantified in Table 1. Removing the gated residual reduces patient-level AUC from 0.798 to 0.548 ($p = 0.009$), the only architectural ablation that reaches significance at this sample size.

Multi-seed comparison of the controlled baselines The variability analysis of Section 5.2 was extended to the three controlled cross-slice baselines, each retrained across three random initializations (seeds 42, 1337 and 2024) on the fixed published split and evaluated at $\tau_{\text{det}} = 0.16$ (Table 10); the proposed model is reported over the same three seeds here, with its full five-seed analysis in Section 5.2. Every seed that ran is reported, without exclusion. Across three seeds, all four architectures fall within a narrow band of patient-level AUC (0.69 to 0.76), and within each architecture no seed differs significantly from another (paired bootstrap, all $p > 0.18$); TomoMamba attains the highest detection rate of the four. The between-architecture differences are small relative to the seed-to-seed spread, so the ordering obtained from any single run should not be over-interpreted. This is the basis on which we decline to rank the cross-slice architectures on classification and rest the architectural

Table 10 Seed variability on the fixed published split, evaluated on the validation partition at $\tau_{\text{det}} = 0.16$. All models are reported over the same three random initializations (seeds 42, 1337 and 2024), computed by the same procedure as Table 6, for a matched comparison; the five-seed TomoMamba analysis is given separately in Table 6. Seeds that ran are reported without exclusion. The single-run values in Table 9 are the seed-42 runs and are therefore higher than these three-seed means.

Model	Patient-level AUC	Detection rate
Slice-independent (+ CenterNet)	0.6875 $\pm$ 0.0872	0.6133 $\pm$ 0.1222
+ BiGRU	0.7617 $\pm$ 0.0476	0.6044 $\pm$ 0.0539
+ Transformer	0.7071 $\pm$ 0.0421	0.6133 $\pm$ 0.0400
TomoMamba (Ours)	0.7029 $\pm$ 0.0819	0.6978 $\pm$ 0.0888

argument on localization at matched false-positive cost and on the significant gated-residual ablation instead.

5.5 Ablation studies

We ablate key design choices within TomoMamba across five categories: loss-function weights, design choices, architectural components, slice selection, and detection threshold (Table 11). All ablations use the same training protocol, data splits, and evaluation pipeline; only the specified component is changed.

Loss-function weights have the largest impact on classification balance. Reducing λ_{cls} from 5.0 to 1.0 drops PatAUC from 0.798 to 0.628, with specificity falling to 0.25, consistent with the dense detection loss dominating without explicit upweighting. Increasing λ_{cls} to 10.0 produces the opposite failure: the model predicts cancer for every patient (Sens = 1.0, Spec = 0.05). Setting all weights to 1.0 (PatAUC = 0.648) markedly underperforms the tuned configuration. Removing the peak regularization loss reduces specificity from 0.70 to 0.40 (J drops from 0.45 to 0.35), so peak regularization matters for suppressing spurious activations that would otherwise mislead the ROI classifier.

Copy-paste augmentation contributes meaningfully to classifier specificity: removing it drops PatAUC from 0.798 to 0.725 and J from 0.45 to 0.25. The 7×7 ROI crop provides the best overall balance. Smaller crops (5×5) achieve a comparable J (0.45) but a much lower detection rate (0.440 vs. 0.720). Larger crops (9×9) degrade both classification (PatAUC = 0.580) and specificity, likely because the expanded context dilutes the lesion-specific signal.

Table 11 Ablation study on TomoMamba design choices. Youden's J ($= \text{Sens} + \text{Spec} - 1$) measures balanced discrimination. All rows use $\tau_{\text{det}} = 0.16$ except in the detection-threshold group

Configuration	PatAUC	Sens	Spec	J	Det. Rate	Acc
Full model ($\tau_{\text{det}} = 0.16$)	0.798	0.75	0.70	0.45	0.720	0.725
<i>Loss function ablations</i>						
$\lambda_{\text{cls}} = 1.0$ (was 5.0)	0.628	0.90	0.25	0.15	0.533	0.575
All $\lambda = 1.0$ (equal weights)	0.648	0.90	0.45	0.35	0.493	0.675
$\lambda_{\text{peak}} = 0.0$ (no peak reg.)	0.773	0.95	0.40	0.35	0.653	0.675
$\lambda_{\text{cls}} = 10.0$	0.590	1.00	0.05	0.05	0.720	0.525
<i>Design choice ablations</i>						
No copy-paste augmentation	0.725	0.85	0.40	0.25	0.653	0.625
ROI crop 5×5 (was 7×7)	0.746	0.95	0.50	0.45	0.440	0.725
ROI crop 9×9 (was 7×7)	0.580	0.80	0.45	0.25	0.480	0.625
<i>Architecture ablations</i>						
No gated residual (3)	0.548	0.55	0.50	0.05	0.747	0.525
No FPN (single-scale features)	0.643	1.00	0.00	0.00	0.000	0.500
<i>Slice-selection ablations</i>						
Random selection	0.683	0.80	0.45	0.25	0.653	0.625
Uniform spacing	0.758	0.90	0.35	0.25	0.560	0.625
Contiguous block	0.703	0.85	0.45	0.30	0.467	0.650
<i>Detection threshold (τ_{det}) ablations</i>						
$\tau_{\text{det}} = 0.10$ (high sensitivity)	0.759	0.85	0.50	0.35	0.840	0.675
$\tau_{\text{det}} = 0.14$	0.803	0.80	0.65	0.45	0.760	0.725
$\tau_{\text{det}} = 0.18$	0.803	0.75	0.70	0.45	0.707	0.725
$\tau_{\text{det}} = 0.22$ (conservative)	0.715	0.60	0.70	0.30	0.573	0.650

Aggregating view-level probabilities by maximum pooling gives a patient-level AUC of 0.7975, against 0.7850 for mean pooling, 0.7675 for the mean of the top two views and 0.6875 for noisy-or. Maximum and mean pooling are not distinguishable ($p = 0.73$), while noisy-or is significantly worse ($p = 0.013$).

Both the gated residual and the FPN are essential. Replacing the learned gate with simple additive fusion reduces PatAUC to 0.548 ($J = 0.05$) while the detection rate holds steady, identifying the gated residual as the component on which classification depends. Removing the FPN eliminates detection entirely (detection rate = 0) and collapses classification into all-cancer predictions.

Slice selection was ablated by rebuilding the dataset with three alternative strategies and retraining under otherwise identical conditions. Each model was evaluated on the validation split built with the rule it was trained on. All four conditions use the same preprocessing script and retain the annotated slice; only the rule filling the remaining fourteen slots changes, so detection is measured against the same ground truth throughout. The fully annotation-independent condition, in which the annotated slice is not retained, is reported separately in Section 5.8. Gradient-based selection matches 72.0% of lesions at 1.47 false positives per volume, against 65.3% at 2.64 for random selection, 56.0% at 2.61 for uniform spacing and 46.7% at 2.72 for a contiguous block. It retains more lesions while producing just over half the false positives of any alternative. Patient-level AUC differences do not reach significance for random ($p = 0.23$), uniform ($p = 0.60$) or contiguous ($p = 0.30$) selection, so we base the justification on lesion retention and false-positive behavior, not on classification accuracy. The contiguous condition is informative in a second respect: it contains no gaps between selected slices by construction, yet yields the lowest detection rate of the four, indicating that the irregular spacing produced by gradient ranking is not detrimental to depth modeling.

PatAUC is relatively stable across $\tau_{\text{det}} \in [0.14, 0.18]$, and classification performance remains identical at $\tau_{\text{det}} = 0.16$ and 0.18. Three thresholds attain the maximum Youden's J of 0.45. We resolve the tie by specificity, then by detection rate: $\tau_{\text{det}} = 0.14$ is eliminated on specificity (0.65 against 0.70), and $\tau_{\text{det}} = 0.16$ is preferred to 0.18 on detection rate (0.720 against 0.707). Specificity is prioritized at this stage because Stage 2 operates diagnostically on cases already triaged as suspicious, so sensitivity is primarily Stage 1's responsibility. This rule was applied to a validation sweep and was not pre-specified.

5.6 Detection analysis

The detection rate reported above counts matched ground-truth lesions, so it describes only one side of detector behavior. A detector can attain a reasonable detection rate while producing many unmatched predictions. We therefore sweep the detection threshold and report the resulting FROC on both partitions (Fig. 7, Table 12).

At the operating threshold $\tau_{\text{det}} = 0.16$, the model produces 110 unmatched detections over 75 validation volumes (1.47 per volume) and 162 over 121 test volumes (1.34 per volume). Sensitivity at fixed false-positive rates is 0.445, 0.650, 0.747 and 0.849 at 0.5, 1, 2 and 4 false positives per volume on validation, and 0.525, 0.579, 0.689 and 0.765 on test.

These figures permit a like-for-like comparison with published detection results, a comparison the summary in Section 5.10 cannot support. The 74.3% sensitivity at two false positives per image reported in [11] compares with 74.7% on validation and 68.9% on test at

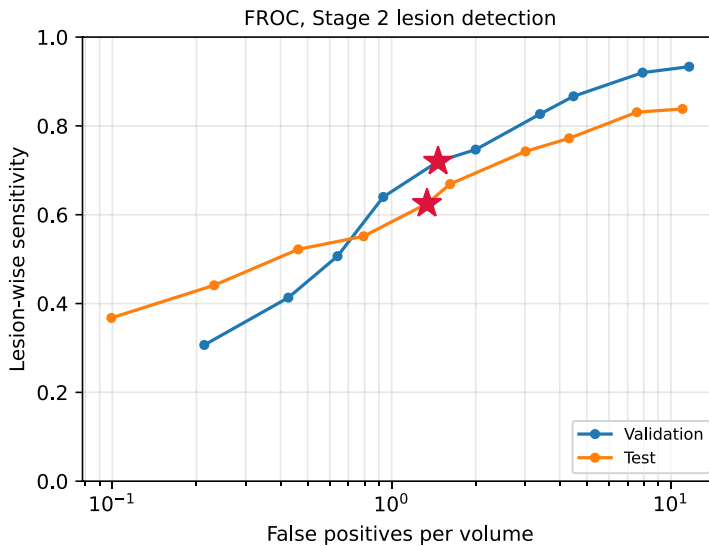


Fig. 7 Free-response operating characteristic for Stage 2 lesion detection on the validation and test partitions. Stars mark the operating threshold $\tau_{\text{det}} = 0.16$. The horizontal axis is logarithmic

Table 12 Detection performance across the threshold sweep. Sensitivity is lesion-wise; false positives are unmatched detections

τ_{det}	Validation			Test		
	Sens	FP/vol	Total FP	Sens	FP/vol	Total FP
0.40	0.307	0.21	16	0.368	0.10	12
0.25	0.507	0.64	48	0.522	0.46	56
0.20	0.640	0.93	70	0.551	0.79	96
0.16	0.720	1.47	110	0.625	1.34	162
0.14	0.747	2.00	150	0.669	1.62	196
0.10	0.827	3.40	255	0.743	3.02	365
0.05	0.920	7.92	594	0.831	7.55	914

two false positives per volume. The 65% at two false positives per breast reported in [4] corresponds to roughly one false positive per volume, since a breast contributes approximately two views; at that rate we obtain 65.0% on validation and 57.9% on test. The published figure is a test-partition result, so the test comparison is the like-for-like one. We state the unit conversion explicitly because the denominators differ across the literature.

Recall at successively stricter IoU criteria is 0.693, 0.600, 0.347 and 0.080 at $\text{IoU} \geq 0.10$, 0.25, 0.50 and 0.75 on validation, with a mean IoU of 0.475 over matched pairs; the corresponding test figures are 0.588, 0.500, 0.338 and 0.088, with a mean IoU of 0.491. The center-distance criterion used for the headline detection rate follows the convention of the DBTex challenge [13], in which the clinical requirement is to direct the reader's attention to the correct location rather than to reproduce box dimensions exactly; we report both criteria so that the difference is visible.

Finally, at low false-positive rates the test curve lies above the validation curve (0.525 against 0.445 at 0.5 false positives per volume), and mean IoU is marginally higher on test. Localization transfers across partitions considerably better than classification does.

5.7 Failure analysis

Aggregate metrics do not indicate which component of the pipeline fails. We separate failures on cancer volumes into three modes: the lesion was not localized; the lesion was localized but the ROI classifier assigned it to the benign class; or no detection was produced at all, so the fallback classifier decided the case.

Of 37 cancer volumes in the validation partition, 18 are classified correctly. Two failures are detection misses, 12 are classification misses, and 5 are fallback cases. The test partition shows the same pattern, with 31 correct, 5 detection misses, 15 classification misses and 9 fallback cases. The dominant failure mode is therefore ROI classification, not localization: in 12 of 19 validation failures and 15 of 29 test failures, the lesion was correctly found and boxed before being misclassified.

We tested the most natural explanation, that missed lesions are smaller than detected ones, and it does not hold. Median lesion area is 0.805% of the image among localized lesions against 0.855% among missed ones on validation, and 1.158% against 1.117% on test. The largest missed lesion in the test partition occupies 9.4% of the image.

The classification misses span cancer probabilities of 0.473 to 0.498 on validation and 0.461 to 0.498 on test, and the benign volumes classified as cancer span 0.504 to 0.547 and 0.511 to 0.739 respectively. On the validation partition no failure occurs with confidence: every misclassification there lies within 0.03 of the decision threshold. The classification misses on the test partition are equally tightly clustered (0.461 to 0.498), while a small number of benign volumes are assigned to the cancer class with more separation, up to 0.739. This is consistent with the narrow predicted-probability range and with the threshold instability described in Section 5.2. Unmatched detections are not concentrated on the harder class, occurring at 1.51 per cancer volume against 1.42 per benign volume on validation and 1.43 against 1.25 on test.

The no-detection fallback When the detector produces no candidate, a global-average-pooling classifier decides the volume. This path is invoked on 10 of 75 validation volumes (13.3%) and 16 of 121 test volumes (13.2%), and invocation is approximately balanced across classes (5 of 37 cancer and 5 of 38 benign volumes on validation). Volume-level AUC on ROI-based volumes is 0.714 on validation and 0.675 on test; on fallback volumes it is 0.520 and 0.413, with a sensitivity of 0.000 on both partitions, so the fallback path assigns essentially every case to the benign class. Excluding fallback views raises patient-level AUC from 0.798 to 0.821 on validation and from 0.618 to 0.658 on test, but leaves one validation patient and four test patients without any prediction, since neither of their views produced a detection. We therefore describe the fallback as a coverage guarantee rather than as a component contributing accuracy, and report both figures so that the trade-off is explicit. The comparison also isolates the ROI-cropping design choice within a single set of weights: the two paths differ only in whether features are pooled globally or cropped at a detected center, and the cropped path is better separated on both partitions.

5.8 Slice selection and annotation independence

Because Stage 2 retains annotated slices during preprocessing (Section 4), we characterize the resulting sequences and test directly whether the model exploits that retention.

Depth structure Selected slices are sorted by original index, and depth order is preserved in 803 of 803 validation volumes. Of consecutive selected slices, 93.5% are directly adjacent in the original volume; the median gap is one slice, with a tail extending to 79. The distribution is bimodal, consistent with a contiguous gradient-selected block together with occasional distant slices. No positional embedding, slice index or relative position is supplied to the cross-slice module at any point, so sequence order is the only depth signal the model receives.

The 728 Normal volumes contain no annotations and are therefore selected by gradient ranking alone. Their gap structure shows the same contiguous-block-with-occasional-jump shape as that of biopsied volumes (94.0% of consecutive slices adjacent against 89.0%), which indicates that large gaps are a property of the gradient selector rather than a signature of annotation retention.

Does the model exploit slice position? The mechanism by which retention could leak information is that the model learns to attend to the sequence position at which the annotated slice tends to appear. Detection rate on volumes where the annotated slice occupies position 0 is 0.739, against 0.690 where it appears elsewhere; on the test partition the corresponding figures are 0.660 and 0.548. Neither difference is significant by two-proportion test ($p = 0.64$ and $p = 0.21$), and the per-position breakdown shows no monotonic trend, with detection at positions 1, 2 and 4 equaling or exceeding detection at position 0 on both partitions.

Annotation-independent selection We additionally rebuilt both partitions using gradient ranking alone, with no retention of annotated slices, and re-evaluated the published model without retraining. Under this procedure the annotated slice is retained incidentally in 0 of 75 validation volumes and 4 of 121 test volumes, so no ground-truth box survives on the validation partition. The gradient and Laplacian criteria favor slices with strong, well-resolved tissue boundaries, which are not in general the single slice on which a radiologist chose to place a box, so the two selection criteria are close to independent and coincidence is rare. Detection under annotation-independent selection is therefore not degraded but undefined: no annotation remains against which a prediction could be matched. Patient-level classification, which does not require the annotated slice in order to be scorable, changes little, falling from 0.798 to 0.748 on validation and from 0.618 to 0.585 on test; the validation difference is not significant (paired DeLong, $p = 0.63$). Had the model been exploiting the annotated slice as a marker, removing it from every volume would be expected to collapse classification, not to leave it statistically unchanged. A lesion is present across many slices while only one carries a radiologist box, so gradient ranking continues to select lesion-bearing tissue even when it does not select the annotated slice.

Because annotations do not exist at deployment, this annotation-independent setting is the one that reflects real-world use. We therefore treat the patient-level classification

obtained here, 0.748 on validation and 0.585 on test, as the deployment-relevant classification result, and the annotation-guided figures reported elsewhere in the paper as an upper reference measured where ground truth is available. The two differ by less than the sampling uncertainty of the validation partition, which is the evidence that the annotation-guided figures are not inflated by slice retention.

5.9 Computational cost

Table 13 reports parameter count, floating-point operations, peak GPU memory and inference latency for all five architectures, measured under identical conditions on an idle NVIDIA RTX 4090 Laptop GPU with an input of $1 \times 15 \times 384^2$. The three cross-slice architectures are matched to within 1% on parameter count, so the comparisons in Table 9 are not confounded by capacity. Among them, TomoMamba requires the fewest floating-point operations at comparable peak memory: 460.9 GFLOPs per volume against 481.8 for the bidirectional GRU and 479.3 for the transformer variant. Latency reflects the sequential nature of the selective scan across the fifteen depth steps: 75.4 ms per volume, against 41.8 for the bidirectional GRU and 59.9 for the transformer variant. We therefore do not claim a current inference-speed advantage: at a depth of only fifteen slices the selective scan is slower than both the recurrent and the attention baselines. The efficiency argument is one of scalability rather than of present latency. The linear $O(D)$ cost of the state-space formulation grows more slowly with depth than the $O(D^2)$ cost of attention, so on deeper acquisitions the comparison would be expected to move in TomoMamba's favor, whereas at the fifteen-slice depth used here that asymptotic advantage has not yet materialized. A breast contributes two views and a patient two breasts, so a complete Stage 2 analysis requires at most four volumes. Every configuration in Table 13 completes a patient study in under half a second on a single consumer GPU and in under 1.3 GB of memory, so all five sit within the latency budget of a reading workflow. Training took approximately five hours for Stage 1 and six hours for Stage 2 on the same GPU. The architectural argument rests on selective, input-dependent propagation, and the linear scaling of the state-space formulation is expected to become material at greater acquisition depth, where the depth sequence exceeds the fifteen slices used here. We did not train a 3D CNN baseline; a fair comparison would require its own architecture and hyperparameter search rather than an undertuned model. We therefore note its absence explicitly, so the memory-versus-accuracy contrast between selective state-space propagation and volumetric convolution should be read as argued here rather than measured. This omission stands alongside the absence of external, cross-institution validation discussed in Section 5.11, and both bound the strength of the architectural claim.

Table 13 Resource cost of each Stage 2 architecture under identical measurement conditions. Input $1 \times 15 \times 384^2$; 50 timed iterations after 10 warm-up iterations on an idle GPU. FLOPs are twice the multiply-accumulate count. Latency is per volume; a patient study comprises at most four volumes

Model	Params	GFLOPs	Peak MB	ms/vol
ResNet-18 Classifier	12.11 M	359.7	1,009	27.3
ResNet-18 + CenterNet	13.33 M	451.7	1,059	32.5
ResNet-18 + BiGRU + CenterNet	13.43 M	481.8	1,296	41.8
ResNet-18 + Transformer + CenterNet	13.50 M	479.3	1,166	59.9
TomoMamba (Ours)	13.47 M	460.9	1,274	75.4

Table 14 Published methods evaluated on the BCS-DBT dataset. Results are not directly comparable owing to differences in task formulation, data splits, external data, and evaluation protocols

Method	Year	Venue	Task	Metric	Value	Key Differences
<i>Detection methods (lesion-level)</i>						
Buda et al. [4]	2021	JAMA NO	Det.	Per-breast sens. @2FP	0.65	Official split
DBTex best ^a [13]	2023	JAMA NO	Det.	Mean sens.	0.957	Ext. data, ensemble
Hossain et al. [11]	2022	Med. Phys.	Det.	Sens. @2FP	0.743	985-pt subset
<i>Classification methods</i>						
Elshazli et al. [8]	2022	Appl. Sci.	3-cls	Accuracy	0.916	600-pt subset ^b
SIFT-DBT [7]	2024	IEEE ISBI	Cls.	Vol. AUC	0.9269	Custom split ^c
3D-GMIC [18]	2024	IEEE TMI	Cls.	Image AUC	0.848	NYU-trained ^d
Terrassin et al. [23]	2024	Deep-Breath	Cls.	AUC	0.90	FFDM-trained ^c
Dual-Net [24]	2025	Front. DH	Cls.	Image AUC	0.822	Image-level
<i>Classification + Detection</i>						
Alashban et al. [1]	2025	DIR	Cls.+Det.	TPR	0.89 ^f	Official split
<i>Our method</i>						
Ours (Stage 1)	2026	MTAP	Screening	AUC	0.970 / 0.970^g	Official split, no ext. data
Ours (Stage 2)			Cls.+Det.	PatAUC	0.80 / 0.62^g	Official split, no ext. data
Ours (end-to-end)			Cascade	Breast AUC	0.93 / 0.92^g	Official split, no ext. data

^a NYU B-Team: 35-model EfficientDet ensemble trained with >1M external mammography images and 400K DBT volumes

^b Custom 80/20 split on a subset of 600 patients (200–600 in different experiments)

^c Uses only the BCS-DBT training partition with a custom 7:1:2 subject-wise split; binary normal vs. abnormal task

^d Trained entirely on 85,526 NYU Langone patients; BCS-DBT used only as an external test set

^e Trained on full-field digital mammograms, not DBT; applied to DBT thick slices at inference

^f Cancer TPR read from the YOLOv5-CBAM confusion matrix, not from FROC analysis; not comparable to the sensitivity@FP metrics above

^g Reported as validation / held-out test partition. Ours are the only rows in this table for which both are given. Lesion sensitivity at matched false-positive rates, which is the basis on which the detection rows above are stated, is reported in Section 5.6

5.10 Context: published methods on BCS-DBT

To place these results in context, Table 14 summarizes published methods evaluated on BCS-DBT. We emphasize that **direct numerical comparison across these methods is not meaningful** because of fundamental differences in experimental setup, as noted in the table footnotes. Since BCS-DBT is the only publicly available DBT dataset with lesion-level annotations, cross-dataset validation is not currently possible for any method in this space.

Methods that use external data achieve much higher performance: the top DBTex result (sensitivity 0.957) used over one million external mammography images, whereas methods

trained solely on BCS-DBT typically reach detection sensitivity of 0.65–0.80 and classification AUC of 0.82–0.93. No prior work performs our exact two-stage task on the official splits without external data, which makes direct comparison infeasible. Cross-paper comparison in DBT is hampered by the same lack of standardized protocols that [26] documented for 2D mammography: differences in preprocessing, splits, and metadata standards introduce biases that compromise comparability.

5.11 Limitations

BCS-DBT is currently the only publicly available DBT dataset that provides biopsy-confirmed pathology labels together with lesion-level bounding-box annotations. Within BCS-DBT, only 101 biopsied patients are available for training, a ceiling that constrains all methods equally. Our framework was designed with this reality in mind: the ROI-based classifier, copy-paste augmentation, and Mamba's parameter-efficient depth modeling all target the small-data regime. To guard against over-interpretation, we evaluate with multiple complementary metrics and draw conclusions from relative comparisons under identical conditions rather than from absolute numbers. Future work on semi-supervised learning, cross-domain transfer from 2D mammography, and multi-institutional data pooling would benefit the entire BCS-DBT research community. With 101 biopsied patients for training and 40 for validation, the study is underpowered for benign-versus-cancer classification: the confidence intervals on every Stage 2 classification metric are wide (Table 5), and we draw no conclusion that rests on a difference smaller than those intervals. The small cohort limits both statistical power and generalizability, and every classification claim in the paper should be read subject to this constraint.

A related challenge is the detection–classification trade-off. The detection loss (dense, pixel-level) can dominate the classification loss (a single volume-level label), potentially degrading classification. Careful loss weighting ($\lambda_{\text{cls}} = 5.0$) and the ROI-based classification design help mitigate this, but the tension remains inherent to any multi-task framework operating on small datasets.

Evaluation is also limited to a single institution and a single scanner. BCS-DBT was acquired entirely on Hologic Selenia Dimensions systems with 1 mm inter-slice spacing, from one patient population. Volumes from GE or Siemens systems would differ in noise characteristics, reconstruction, tissue contrast and slice thickness, any of which could affect both the gradient-based slice selection and the depth sequences the model receives. Because BCS-DBT is the only public DBT resource with lesion-level bounding-box annotations and all other public mammography datasets are two-dimensional, this constraint applies to every published DBT detection and classification method, none of which has demonstrated transfer across sites. Establishing that transfer is a prerequisite for clinical claims and is not addressed here. Multi-center validation and scanner-specific adaptation are important future directions.

Several design choices were selected on the validation partition, including the screening and detection thresholds, the checkpoint, the ROI size, the loss weights and the view-aggregation rule. The five-seed analysis in Section 5.2 quantifies the consequence: the five-seed mean predicts the held-out test result more closely than the published peak does. Readers should treat the validation figures as upper estimates. The controlled baselines are now reported across three random initializations, matched to the proposed model on the

same seeds (Section 5.4); the reproducibility comparison across architectures nonetheless remains bounded by the 40-patient validation cohort, on which run-to-run variability and between-architecture differences are of similar size. We therefore report the spread and decline to rank the cross-slice architectures on classification.

Finally, the 40-patient validation set limits the precision of every estimate reported on it: a change of one or two patients shifts sensitivity and specificity by 0.025 to 0.05. This is why we examine several metrics rather than any single number, and why we treat relative comparisons under identical conditions (Table 9) as more informative than absolute values. On the test partition the patient-level classification interval includes 0.50, so the benign-versus-cancer task remains open at this sample size, and Stage 2's demonstrated contribution is lesion localization within a screening cascade.

6 Conclusion

We presented TomoMamba, a two-stage framework whose cross-slice module is, to the best of the authors' knowledge, the first application of selective state-space models to inter-slice feature propagation in DBT. Existing DBT methods either ignore the depth axis or model it with fixed recurrence or attention; Mamba conditions its state transitions on slice content, and the gated residual ensures that this context sharpens rather than dilutes the per-slice signal.

Stage 1 screens each breast study (up to four views per patient) with a breast-level AUC of 0.9696 on the validation partition, filtering out the overwhelming normal majority before expensive analysis begins. Stage 2 operates on the remaining suspicious cases, almost all of which involve a single biopsied breast, so that its breast-level and patient-level metrics coincide up to one validation patient. On the validation partition, it attains PatAUC = 0.80, sensitivity = 0.75 and specificity = 0.70, with lesion localization at $R@0.25 = 0.60$ and a 72.0% detection rate. On the held-out test partition the corresponding figures are 0.62 and 62.5%. All of this is achieved with 200 training volumes and no external data. The strong end-to-end cascade figure (breast-level AUC 0.92 on test) is a property of the screening cascade, driven by the normal-versus-suspicious separation of Stage 1, and not a benign-versus-cancer result; the latter is the more modest Stage 2 figure (patient-level AUC 0.62 on test), and we frame the contribution accordingly.

Three findings stand out from the controlled baselines. First, joint detection grounds the classifier spatially, rescuing it from the instability of global pooling. Second, at matched false-positive cost selective state-space propagation localizes more lesions than every alternative on the validation partition, although the cross-slice architectures as a class are not separable by patient-level AUC. Third, the gated cross-slice residual is the one architectural component whose removal produces a statistically significant loss (0.25 AUC, $p = 0.009$). On the untouched test partition TomoMamba also attains the highest patient-level AUC and the lowest Brier score of the five configurations, although no pairwise difference on that partition reaches significance.

These results are obtained with 200 biopsied training volumes on the only public DBT dataset with lesion-level annotations. We therefore present TomoMamba as a first controlled evaluation of state-space cross-slice propagation in this setting, with the expectation that the approach will require re-evaluation as larger annotated DBT cohorts become available.

Since BCS-DBT remains the only publicly available DBT dataset with lesion-level annotations, future work will target the data bottleneck through semi-supervised learning and cross-domain transfer from 2D mammography, and will pursue validation on additional DBT datasets as they become available. Reader studies and richer intermediate outputs (structured lesion characterization, BI-RADS descriptors, and the sequential reasoning advocated by [27]) are natural next steps.

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Author Contributions **Shahd Soliman:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Yalda Zafari:** Methodology, Writing – review and editing. **Essam A. Rashed:** Writing – review and editing, Supervision. **Mohamed Mabrok:** Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review and editing.

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Data Availability The BCS-DBT dataset is publicly available at <https://doi.org/10.7937/e4wt-cd02>.

Code availability A public repository accompanies this submission: <https://github.com/Minds-R-Lab/TomoMamba>. It contains the patient-level split files, the preprocessing and gradient-based slice-selection scripts, the copy-paste augmentation implementation, the model configurations for every architecture reported, and the threshold-selection and evaluation code. It also contains the per-case prediction dumps underlying every reported table and the random seeds for all reported runs, which together with the configurations are sufficient to reproduce the trained checkpoints.

Declarations

Competing interests The authors declare no competing interests.

Ethical Approval This study uses a publicly available, de-identified dataset (BCS-DBT) from The Cancer Imaging Archive. No additional ethics approval was required.

Consent to participate Not applicable. This study uses a publicly available, fully de-identified retrospective dataset.

Consent to publish Not applicable. No identifiable individual data are reported.

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